

Journal: Ecological Applications

Manuscript type: Method

Smart environmental DNA sampling strategies: a spatial autocorrelation method and interactive planner for survey design

Running headline: Efficient eDNA survey design

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Open Research statement: Data are archived in Zenodo (Guri, 2026; <https://doi.org/10.5281/zenodo.21141367>). Novel code (R and Stan scripts that reproduce all analyses) and the eDNA Survey Planner are provided for peer review in a public GitHub repository (<https://github.com/gledguri/Smart-eDNA-sampling>), and the planner can be used directly in a web browser at <https://gledguri.github.io/Smart-eDNA-sampling/>. Upon acceptance, the version of record of the code and the planner will be permanently archived in Zenodo.

Key words: biodiversity monitoring; decision-support tool; environmental DNA; fisheries management; Gaussian process; metabarcoding; sampling design; spatial autocorrelation; survey cost-effectiveness

Abstract

Environmental DNA (eDNA) surveys are increasingly used to monitor marine biodiversity, yet managers who must budget and justify surveys have little quantitative guidance on how many samples to collect, and because species differ in patchiness, no single sample size serves all. We hypothesized that a taxon's spatial autocorrelation length-scale (ρ), which describes how quickly its eDNA field changes across space, can guide this decision, because smoothly varying fields can be reconstructed from fewer samples than patchy ones. To test this, we simulated Gaussian process (GP) fields spanning a wide range of ρ , fitted Bayesian GP models to learn their parameters, progressively removed samples and refitted the models, and compared the resulting predictions and parameter estimates with those from the complete fields. The resulting prediction errors are summarized in an invertible error–effort equation that converts a target prediction error and a prior estimate of ρ into a minimum sampling density and is implemented in a free, browser-based survey planner. We then applied the method in a case study combining eDNA data for 12 fish species (358 samples from five depth strata along the U.S. West Coast) with 200 simulated fields ($\rho = 25$ –1000 km). Marginal gains in prediction accuracy became negligible beyond 4–5 samples per 10,000 km² per depth layer for smoothly distributed species ($\rho > 500$ km), whereas for patchily distributed species ($\rho < 50$ km) they became negligible only beyond 8–11, more than twice the sampling density. Error patterns in the simulations closely matched those in the observed data. Together, these results show that the sampling effort needed to characterize a species' distribution is not universal but depends on its spatial autocorrelation, and can therefore be specified in advance. The planner lets survey designers set defensible, species-specific effort before a cruise.

Introduction

Biodiversity monitoring programmes underpin decisions about where to fish, what to protect and how to report progress against national and international conservation targets, yet the budgets available to them are effectively fixed. How many samples to collect, and where, is therefore a management decision as much as a statistical one (Yoccoz et al., 2001). Survey design must balance the need for sufficient data for detecting trends and changes in population while also minimizing cost (i.e., maximizing sampling efficiency) (Stein & Ettema, 2003; Xu et al., 2015; Zhao et al., 2022).

The emergence of environmental DNA (eDNA) sampling has revolutionized monitoring capabilities, offering a non-invasive method to detect species presence and estimate relative abundance (Gold et al., 2021; Shelton et al., 2022; Taberlet et al., 2012). Recent synthesis work shows that eDNA methods have matured sufficiently for key uncertainties to be quantified and incorporated into existing fisheries stock-assessment frameworks, enabling the development of cost-effective abundance indices that complement conventional surveys (Baetscher et al., 2025; Guri et al., 2024; Kirtane et al., 2021; Salter et al., 2026). Because agencies are now evaluating eDNA as an operational component of biodiversity assessment rather than as an exploratory tool, explicit and defensible guidance on how much to sample has become a prerequisite for uptake rather than a refinement of it.

While increased sampling intensity generally reduces variance in population estimates and improves the precision of spatial pattern detection in species distribution models (Schweiger et al., 2016), this relationship exhibits diminishing returns where beyond a certain threshold, additional

sampling effort produces marginal gains in statistical precision that may not justify the associated costs and logistical demands (Perret et al., 2022; Zhao et al., 2022).

We conceptualize species spatial distributions as realizations of a Gaussian Process (GP), in which spatial dependence is characterized by a species-specific length-scale parameter (ρ) describing the distance over which observations remain correlated. A species with a long length-scale varies gradually across space and can therefore be reconstructed from fewer samples, whereas a species with a short length-scale exhibits more rapid spatial variation and consequently requires denser sampling (Figure 1). Thus, knowledge of species-specific spatial autocorrelation can provide a quantitative basis for informing sampling effort (Fortin et al., 1989). Yet, although general design guidance exists, for example on spatially balanced allocation of samples (Perret et al., 2022), on designs for estimating spatial autocorrelation (Laroche, 2024) and on eDNA sampling designs for fish diversity in lakes (García-Machado et al., 2024), there is still no quantitative, transferable procedure that returns how many eDNA samples a survey needs to reach a given precision, and design guidance is rarely used unless it is accessible to those who plan surveys. Here we present a quantitative, simulation-based method, together with a free, interactive planner that implements it, built on the premise that the sampling density needed to reconstruct a species' distribution to a given accuracy can be predicted from its spatial autocorrelation length-scale. Using the simulations, we quantify how spatial autocorrelation and sampling effort affect (i) the predictive accuracy of species distribution models and (ii) the accuracy of GP parameter estimation. We then test whether the error structure of the simulations holds in real eDNA fields, using data for 12 fish species from the California Current, and use the same data to provide reference length-scales and to show how the planner can be used to judge whether an existing

survey was large enough. Our objective is deliberately operational, namely to provide survey planners with guidance that can be applied before a cruise, based on a prior estimate of the spatial autocorrelation of target taxa and the level of precision required. We describe the underlying method and the planner, evaluate them on simulated and empirical data, and outline applications to survey planning, monitoring and management.

Description of the method

The method links sampling effort to the accuracy of spatial inference within a common GP framework and proceeds in four steps, Step 0 to Step 3 (Figure 2): (0) generate simulated eDNA fields from a GP model across a wide range of spatial autocorrelation length-scales, so that the spatial structure of each field is known by construction and effort can be varied over a range no field campaign can achieve; (1) fit the same GP model to each complete simulated dataset to learn its governing parameters and to predict a reference field across a fixed prediction grid; (2) remove samples at progressively lower sampling densities and refit the GP model to each reduced-effort dataset and compare the parameters and the predicted fields with those obtained in Step 1, quantifying how parameter recovery and prediction accuracy degrade as effort is reduced; and (3) summarize the resulting prediction error as a function of sampling density and spatial autocorrelation in a regression that can be inverted to return the sampling density required for a user-defined error tolerance. The resulting equation is implemented in the eDNA Survey Planner, so that end users need only a prior estimate of spatial autocorrelation and a target error. The same workflow is also applied to observed eDNA data to benchmark the simulations against real fields. We describe each step generically, giving the values used in the case study where relevant. The results of the simulations are reported first (Method performance), followed by the empirical case

study, which checks the simulations against real fields and demonstrates the planner on an existing survey.

Modelling framework: Gaussian processes

The GP approach is grounded in the assumption that species distributions and abundances change in a smooth and continuous manner across space (or other covariates), reflecting correlated responses to spatial, environmental, and biological covariates (Lewy & Kristensen, 2009). At the core, in simple terms, GPs posit that observations made at proximate locations or covariate values should be similar (Dormann et al., 2007). We assert that many species can be well described using relatively simple covariance models (Thorson et al., 2015). Specifically, we use an isotropic, squared exponential covariance model for the distribution of a species. Let $X = \{x_1, x_2, \dots, x_N\}$ be the set of N observed spatial locations. Then species (eDNA) abundance or density Y across all sampled locations (X) is modeled as,

$$Y \sim MVN(\mu, \mathbf{K} + \sigma^2 \mathbf{I}), \quad 1$$

where μ is the global mean abundance, σ^2 is the non-spatial variance, $\mathbf{I}$ is an $n \times n$ identity matrix and $\mathbf{K}(X, X)$ is the covariance matrix whose $(i, j)^{\text{th}}$ entry is given by the squared exponential kernel function,

$$\mathbf{K}(X, X) = k(x_i, x_j) = \alpha^2 \exp\left(-\frac{(x_i - x_j)^2}{2\rho^2}\right) \quad 2$$

where α^2 is the amplitude (i.e., the marginal variance of the spatial process, controlling the overall magnitude of variation in abundance), ρ is the length-scale parameter (in $x_i - x_j$ units; hereafter spatial autocorrelation).

Step 0: Generating the simulated spatial field

We generated in-silico eDNA concentration data using the GP model (Eq. 1-2) across the same spatial domain and prediction grid (Q) used in the empirical analysis. We defined this grid (Q) as 18,916 discrete point estimates, representing the centroids of approximately 8×8 km spatial cells distributed across five depth layers. The number of points differed among depth layers (4,000, 4,000, 4,000, 3,622, and 3,294 points, respectively) because deeper layers covered a smaller spatial extent than shallower layers due to bathymetry.

The simulations were designed to explore a wide range of spatial structures by varying the GP length-scale parameter (ρ) across ten levels: 25, 50, 75, 100, 200, 300, 400, 500, 800, and 1000 km (Appendix S1: Table S1). For each value of ρ we generated (arbitrarily) 20 independent replicate spatial fields, resulting in a total of 200 simulated datasets (10 ρ levels $\times$ 20 replicates; Appendix S1: Table S1). Similar to GP inference of the empirical data, within each simulation, ρ was held constant across depth layers, but each layer had a separate mean (μ_d) drawn randomly from a normal distribution with mean of 0 and a standard deviation of 3. We focused only on spatial autocorrelation across latitude and longitude, allowing the eDNA concentration at each depth to be independent. The amplitude (α^2) and the non-spatial variance (σ^2) in Eq. 1 and 2 were fixed at 4 and 3, respectively, to mirror the empirical estimates.

Given the large number of projected eDNA points (Q), fitting a GP model using all locations ($\approx$ 20,000 input values) was computationally prohibitive. We therefore drew a subset of 4,000 points from each of the 200 simulated datasets as the “complete” observation dataset (Y) and its covariate (X), which constitutes the data used for all downstream analyses.

160

161 **Step 1: Estimating taxon-specific spatial structure**

162 Given simulated log eDNA concentration values (Y) paired with spatial coordinates (X) for each
163 sampling location (x ; $N = 4,000$ per simulated field), we used the GP model (Eq. 1–2) to infer the
164 underlying spatial structure and learn key parameters governing the spatial field, including the
165 depth-specific mean (μ_d), spatial autocorrelation (ρ), amplitude (α^2), and non-spatial variance (σ^2).
166 To predict the eDNA concentrations (Z) across the spatial prediction grid (Q) for a given simulated
167 field, we compute the covariance matrix among observed locations, $\mathbf{K}(X, X')$, using the squared
168 exponential kernel and the non-spatial variance term ($\sigma^2 \mathbf{I}$). Using the Cholesky decomposition we
169 solved for $(\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1}(Y - \mu_d)$ and we combined it with the cross-covariance between observed
170 and prediction points, $\mathbf{K}(X, Q)$, to yield the predictions (Z) at each spatial point (Q):

$$171 \quad Z = \mu_d + \mathbf{K}(X, Q)^\top (\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1} (Y - \mu_d) \quad 3$$

172 The resulting predicted field (Z) served as the “true” spatial field against which all subsequent
173 reduced-effort predictions were compared (Figure 2A).

174

175 **Step 2: Quantifying information loss under reduced sampling effort**

176 To evaluate how sampling effort (E) influences spatial reconstruction and prediction
177 accuracy, we generated a series of reduced-effort observation datasets (Y_E coupled with X_E) by
178 randomly thinning the observation set (Y) and its spatial covariate (X) to fixed sample sizes
179 (Figure 2). Each reduced-effort observation dataset (Y_E) together with its covariate (X_E) was refit
180 with the same GP model (Eq 1-2) and produced effort-specific spatial predictions (S_E) on the full
181 grid Q , by substituting S_E for Z , Y_E for Y and X_E for X in Eq. 3. This framework enables direct,

like-for-like comparisons of reconstructed fields across sampling intensities since the predicted field is the same (Q). Sampling effort (E) was quantified as sampling density:

$$E = \frac{N}{A \times H} \times 10000 \quad 4$$

where A is the spatial area of the study domain (200,000 km²), and H is the number of strata, here depth layers (d ; $H = 5$), yielding E in units of samples per 10,000 km² per depth layer.

For each model prediction, we calculated the prediction error (ε), defined as the root mean squared error between spatial fields predicted from the full dataset (Z) and those predicted from each reduced sampling scenario (S_E) as follows (Figure 2):

$$\varepsilon_{\text{sim}} | E = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_i - S_{iE})^2} \quad 5$$

with n being the total number of Q grid points ($n = 18,916$) where the predictions (Z and S_E) are projected. Here, $\varepsilon_{\text{sim}} | E$ denotes the prediction error of the simulated fields at sampling density E , and $\varepsilon_{\text{obs}} | E$ the prediction error of the observed data at sampling density E ; hereafter we write ε_{sim} and ε_{obs} .

Additionally, to evaluate the influence of sampling effort on GP parameter estimation, we compared the estimated parameters ($\rho, \mu_d, \sigma^2, \alpha^2$) from effort-specific datasets (Y_E) and covariates (X_E) to the true values used to generate the simulated fields. Differences between estimated and simulated parameter values were quantified using the root mean squared error (hereafter τ).

Step 3: Solving for the sampling density

To examine how sampling effort and spatial autocorrelation jointly shape prediction error, we fit a log-linear model to the simulated prediction errors (ε), isolating each factor's contribution.

We modeled prediction error (ε) as a function of sampling density (E) and spatial autocorrelation (ρ), including their interaction using a linear model (*lm* function) in R as follows:

$$\log(\varepsilon) = \omega + \beta \log(E) + \theta \log(\rho) + \gamma \log(E)\log(\rho) \quad 6$$

where ω is the intercept, β slope of sampling density (E), θ is the slope of spatial autocorrelation (ρ) and γ is the coefficient on the interaction between the two latter factors.

For a given ρ and a stated tolerance for prediction error (ε^* , the level of error the manager is willing to accept), Eq. 6 reduces to a power law in E that can be inverted to return the minimum sampling density (E^*) required to achieve that target error:

$$E^* = \exp\left(\frac{\log(\varepsilon^*) - \omega - \theta \log(\rho)}{\beta + \gamma \log(\rho)}\right) \quad 7$$

Eq. 7 is deterministic: it returns a single sampling density for a given ρ and ε^* , and it describes the expected relationship between effort, spatial autocorrelation and error rather than the scatter around it. Realized errors in any individual survey will therefore vary around ε^* , and the range of sampling densities compatible with a given target is wider than the single value returned. However, here we focus only on a single value, which is based on the mean. The same relationship can also be run in reverse, so that a fixed budget of N_H samples per stratum supports a survey area of 10,000 N_H/E^* at the target error. Eq. 7 is calibrated over $\rho = 25\text{--}1000$ km and $E = 1.5\text{--}20$ samples per 10,000 km² per stratum, and results outside this range are extrapolations.

All the GP models (equation 1 and 2) were fitted in a Bayesian framework using Stan probabilistic programming language (Stan Development Team, 2023)10/2/26 9:14:00 PM, with weakly informative priors, and the model in Eq. 6 was fitted using the *lm()* function in R (R Core Team,

2024); sampler settings, convergence diagnostics and prior distributions are given in Appendix S1:
Section S3 and Table S2.

Method performance: simulation results

We evaluated the method on the 200 simulated fields generated in Step 0. Here we report how the prediction error and the recovery of the Gaussian process parameters change with sampling density and spatial autocorrelation, and how the prediction error is summarized in the error–effort model (Eq. 6) that underlies the planner.

Prediction error and the error–effort model

Prediction error in simulation data (ε_{sim}) decreased non-linearly with increasing sampling effort and with higher spatial autocorrelation (Figure 3A). These patterns were supported by a log–log linear model where the prediction error (ε) decreased exponentially for every exponential increase in sampling density ($\beta = -0.176$) and spatial autocorrelation ($\theta = -0.228$), with an additional negative interaction ($\gamma = -0.0684$) and with an intercept of $\omega = 1.65$. For instance, at a sampling intensity of 6 samples per 10,000 km², species with spatial autocorrelation values of 50 km had mean prediction errors of 1 (Figure 3A) indicating that on average every point predicted the eDNA concentration will differ from the predicted true concentration (Z) by a factor of 2.72 (for perspective a concentration of 100 copies/L will be predicted between 37-270 copies/L; Appendix S1: Figure S3). In contrast, for the same sampling effort (i.e., 6 samples per 10,000 km²) but species with spatial autocorrelation $\rho > 500$ km had errors below 0.4 indicating an average of over- and underestimation by a factor of 1.5 (for perspective a concentration of 100 copies/L will be predicted between 67-150 copies/L; Appendix S1: Figure S3). When quantifying the marginal

reduction in prediction error ($\Delta\epsilon_{sim}$) across successive increases in sampling density (ΔE), improvements were greatest at low sampling effort and declined rapidly with additional sampling, with $\Delta\epsilon_{sim}$ approaching zero at higher sampling densities (Figure 3B). Marginal gains effectively plateaued when sampling effort increased above 10 samples per 10,000 km², depending on the ρ (Figure 3B). Importantly, diagnostic checks revealed no systematic bias in the predictions, so the observed variation reflects pure precision (Appendix S1: Figure S4). For interpretative purposes, we adopt an arbitrary threshold of $\Delta\epsilon_{sim}/\Delta E = -0.05$ to denote negligible marginal gains. However, this value is arbitrary and may be redefined according to study-specific accuracy requirements.

GP parameter recovery

Sampling effort and spatial autocorrelation strongly influenced the ability of the GP model to recover the parameters used to generate the simulated spatial fields (Figure 4). Estimation error for the depth-specific mean (μ_d) decreased with increasing sampling density for simulated fields with small spatial autocorrelation but showed the opposite pattern for fields with large spatial ranges (Figure 4C). Specifically, τ (root mean squared error between estimated and simulated) for μ_d increased with sampling effort when spatial autocorrelation exceeded 400 km, indicating that mean estimates became increasingly biased in highly spatially structured fields. Additionally, the recovery of the spatial autocorrelation length-scale (ρ) was highly sensitive to sampling density (Figure 4A). At low sampling effort ($E < 4$), estimates of ρ showed substantial error, particularly for simulations with large spatial ranges. However, estimation accuracy improved rapidly as sampling effort increased, with τ declining sharply and stabilizing once sampling densities were higher than 6 samples per 10,000 km², beyond which additional sampling produced only marginal improvements in ρ estimation.

Estimation error for the amplitude parameter (α^2) declined with increasing sampling density for simulations with small spatial autocorrelation, while remaining relatively stable across sampling densities for simulations with larger spatial ranges (Figure 4D). Overall improvements in α^2 estimation were therefore limited once sampling effort increased above 6 samples per 10,000 km². The non-spatial variance parameter (σ^2) was consistently well recovered across sampling densities (Figure 4B). τ declined sharply with increasing sampling density and approached near-zero values once sampling densities exceeded 6 samples per 10,000 km².

Implementation: the eDNA Survey Planner

To make Eq. 6–7 usable without statistical software, we implemented them in the eDNA Survey Planner, a self-contained browser application (a single HTML file) that runs offline, requires no installation and uploads no data (Appendix S1: Figure S6; Open Research statement). It answers the two questions survey designers most often face. “How many samples?” takes a target error (ε^*), a prior estimate of ρ , the survey area per stratum and the number of strata, and returns the required sampling density, the samples per stratum and the total site–stratum observations. “How much area?” takes ε^* , ρ and a sample budget per stratum, and returns the area that budget can support at the target error.

Because ε is a root mean squared error on the natural-log scale, the planner translates it into a concentration range: around a reference concentration C , predictions typically fall within $\exp(\log(C) \pm \varepsilon)$, a multiplicative factor of e^ε (e.g., $\varepsilon = 0.8$ corresponds to about 45–223 copies/L around a true 100 copies/L; Appendix S1: Figure S3). An interactive error–effort curve shows where the target lies on the curve of diminishing returns, and a one-dimensional GP simulation

displays a true field, the samples required to reach ε^* and the resulting reconstruction, alongside one quarter of that effort. Spatial autocorrelation can be entered directly or chosen from presets (25–1,000 km).

Default coefficients are those of Eq. 6 estimated from the simulations, but they can be edited, so that programs can substitute a calibration for their own region, taxa or assay by applying Steps 1–4 with the archived code. Results outside the calibrated range are calculated but flagged as extrapolations, and the planner states that ε is an average error across the survey domain rather than a pointwise confidence interval. Recommendations follow the central fitted curve; uncertainty in the coefficients is not yet propagated.

Case study: a multispecies eDNA survey of the California Current

The case study serves three purposes: it tests whether the simulation-based error model holds in real eDNA fields, where the true field is unknown; it provides reference length-scales for 12 fish taxa that planners can use as priors; and it shows how the planner can be used to judge whether an existing survey was sufficient enough.

Study system and eDNA data

Environmental DNA samples were collected during the Joint U.S.-Canada Integrated Ecosystem and Pacific Hake Acoustic-Trawl Survey aboard the NOAA ship *Bell M. Shimada* in summer 2019, from central California to northern Washington (38.36°–48.46° N, 123.53°–125.93° W; study domain of approximately 200,000 km²). Seawater was collected with Niskin bottles at five target depths (0, 50, 150, 300 and 500 m) and filtered onboard, yielding 358 site–depth samples

from 152 sampling locations. Species-specific eDNA concentrations (log copies/L) were estimated for 12 fish species by jointly modeling 12S rRNA metabarcoding reads (MiFish primers), a mock community to correct for amplification bias and qPCR measurements of Pacific hake (Guri et al., 2024, 2025). The focal assemblage comprised pelagic, mesopelagic and demersal taxa characteristic of the California Current ecosystem: *Clupea pallasii*, *Engraulis mordax*, *Leuroglossus stilbius*, *Microstomus pacificus*, *Sardinops sagax*, *Scomber japonicus*, *Sebastes entomelas*, *Stenobranchius leucopsarus*, *Tactostoma macropus*, *Tarletonbeania crenularis*, *Thaleichthys pacificus* and *Trachurus symmetricus*. Sampling, laboratory and bioinformatic details are given in Appendix S1: Section S1.

Spatial structure of the focal species (Step 1)

As in Step 1 of the simulations, we fitted the GP model (Eq. 1–2) to the log eDNA concentrations of each of the 12 species ($N = 358$ site–depth samples) and used Eq. 3 to predict eDNA concentrations across the prediction grid (Q) defined in Step 0.

Across species, depth-specific global mean eDNA concentrations generally declined with increasing sampling depth (Figure 5). Species-specific length-scale estimates varied considerably across species (Figure 5A), likely reflecting differences in habitat use and movement behavior. Surface-dwelling species tended toward shorter spatial autocorrelation lengths and showed more pronounced differences between surface and deep mean eDNA concentrations (μ ; Figure 5B), though exceptions were evident. Deep-dwelling species broadly exhibited longer spatial autocorrelations (Figure 5A) and relatively stable mean eDNA concentrations across depth strata, although this pattern was not universal. *Tactostoma macropus*, for instance, showed a

comparatively short length-scale despite its deep-water affinity. Estimated length-scales ranged from 22 to 1283 km across the 12 species (Table 1).

Information loss in the observed data (Step 2)

As in Step 2, we randomly thinned the 358 observations to $N = 330, 300, 260, 220$ and 150 (Figure 2B), refit the GP model to each reduced dataset and calculated the prediction error (ϵ_{obs}) against the field predicted from the full dataset as in Eq. 5, repeating the procedure across 20 random realizations for each thinning intensity.

Prediction error in empirical data decreased with increasing sampling density across all species (Appendix S1: Figure S1 and S2). However, this reduction was strongly modulated by the degree of spatial autocorrelation (Appendix S1: Figure S2). Species characterized by short spatial autocorrelation (small ρ values) consistently exhibited higher prediction error (ϵ_{obs}) across all sampling densities, whereas species with wider spatial autocorrelation structures (e.g., $\rho > 500$ km) showed substantially lower prediction errors (ϵ_{obs}) even under comparatively sparse sampling regimes.

Agreement between simulated and observed error models

Because the empirical benchmark (ϵ_{obs}) was defined by the 358 observed samples, whereas the simulation benchmark (ϵ_{sim}) used 4,000 samples, the two errors are not directly comparable (Figure 2). We therefore repeated the thinning on each simulated field using its $N = 358$ subsample as the reference ($\epsilon_{sim,358}$), fitted Eq. 6 to $\epsilon_{sim,358}$ and to ϵ_{obs} , and compared the resulting coefficients to

test whether the simulations reproduce the error structure of real eDNA fields (Appendix S1: Section S2).

Prediction error (ϵ) followed similar patterns using both empirical and simulation datasets. Specifically, the estimated intercept (ω), the influence of sampling effort (β), the effect of spatial autocorrelation (θ), and their interaction term (γ) all showed consistent directional effects and aligned closely with the 1:1 line, demonstrating a high degree of structural agreement between the empirical and simulated error models (Appendix S1: Figure S1).

Applying the planner

To show how the planner is used, we asked whether the 358 site–depth samples collected in 2019 ($E = 3.6$ samples per 10,000 km² per depth layer) were enough to reach a target prediction error of $\epsilon^* = 0.5$, corresponding to predictions within a typical factor of about 1.65 of the reference field. For each species we entered its estimated length-scale (Step 1 applied to the observed data), the survey area (200,000 km² per layer) and the five depth layers into Eq. 7, which returns the required sampling density (E^*) and the corresponding total number of samples (Table 1). The samples collected were sufficient for the two species with the longest length-scales, *Leuroglossus stilbius* and *Microstomus pacificus*, for which the planner required 291–295 samples, about 17–19% fewer than were collected, and were close to the requirement for *Sebastes entomelas* (373 samples; $\rho = 763$ km). For the remaining 9 species the planner required 2.2 to 19.8 times as many samples as were collected, and the requirement rose steeply as the length-scale decreased (for example 1,011 samples for *Clupea pallasii*, $\rho = 150$ km, and 7,078 for *Tarletonbeania crenularis*, $\rho = 22$ km). Because 5 species have length-scales outside the calibrated range ($\rho = 25$ –1000 km) and 4 require

more than the calibrated maximum of 20 samples per 10,000 km² per layer (Table 1), the largest requirements are extrapolations and are best read as showing that the target was not attainable with the 2019 effort for those species, not as precise sample sizes.

Table 1. Reference spatial autocorrelation length-scales (ρ) for 12 fish species in the California Current, and the sampling required to reach a target prediction error of $\epsilon^* = 0.5$ (Eq. 7; 200,000 km² per layer, five depth layers). E^* is the required sampling density (samples per 10,000 km² per depth layer), N^* the total number of site–depth samples required ($E^* \times 100$), and the last column compares N^* with the 358 samples collected (values below 1 indicate that the collected samples exceeded the requirement). † Outside the calibrated range of the error–effort model ($\rho = 25$ –1000 km or $E = 1.5$ –20 samples per 10,000 km² per layer): extrapolation.

Species	ρ (km)	E^*	N^*	$N^* / 358$
Leuroglossus stilbius†	1283	2.9	291	0.81
Microstomus pacificus†	1243	3.0	295	0.83
Sebastes entomelas	763	3.7	373	1.04
Engraulis mordax	207	8.0	804	2.25
Sardinops sagax	181	8.8	882	2.46
Clupea pallasii	150	10.1	1,011	2.82
Scomber japonicus	89	15.4	1,540	4.30
Tactostoma macropus	79	17.0	1,702	4.75
Stenobranchius leucopsarus†	41	32.8	3,283	9.17
Trachurus symmetricus†	24	60.8	6,076	17.0
Thaleichthys pacificus†	23	65.3	6,532	18.2

Tarletonbeania crenularis†	22	70.8	7,078	19.8
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Discussion

We present a method, and a planner that implements it, for translating species-specific spatial autocorrelation into the sampling effort required by eDNA spatial surveys. ~~The case study shows~~ The simulations and the case study show that optimal sampling effort is not universal but depends critically on both the spatial autocorrelation of target species and the specific inferential objectives of the survey.

~~Diminishing returns~~ Method performance: diminishing returns of sampling effort on prediction accuracy

The value of additional sampling effort is not fixed but depends on the spatial structure of the target species (Xu et al., 2015; Zhao et al., 2022). Prediction error decreased non-linearly with sampling effort and with increasing spatial autocorrelation, confirming that homogeneously distributed species are inherently easier to reconstruct than patchier ones at equivalent effort (Wagner & Fortin, 2005). This interaction matters because it shows the value of additional samples is not universal, where the same increase in effort can produce substantial gains for one species and only marginal gains for another (García-Machado et al., 2024). Our results move beyond a general statement of sampling efficiency, providing a quantitative framework for understanding when additional effort is informative and when it yields diminishing returns.

Our adopted threshold for negligible marginal gains ($\Delta\epsilon_{sim} = -0.05$) serves as a practical heuristic, but acceptable error is inherently context dependent. For instance, studies requiring high-resolution quantitative estimates might justify high sampling effort to achieve stricter precision, while, broad-scale biodiversity screenings might tolerate looser precision, allowing for even sparser sampling

(Perkins et al., 2021; Radinger et al., 2019). Moreover, the species spatial autocorrelation parameter plays an additional role in defining the decision on sampling effort. Highly autocorrelated species (e.g., $\rho > 500$ km) reach the precision threshold of negligible improvement at much lower sampling densities (roughly 4 to 5 samples per 10,000 km²), conversely, patchily distributed species ($\rho < 50$ km) require a denser sampling network (roughly 8 to 11 samples per 10,000 km²) before the marginal gains drop below the threshold. Once the required precision and ρ are known (from prior literature, pilot sampling or expert judgement), the fitted log–log model (Eq. 6; Figure 43B; Appendix S1: Figure S5) can be inverted (Eq. 7) to solve for the minimum sampling density E required to reach a target error, a calculation the planner performs interactively.

Although we report effort per 10,000 km², the underlying point is simpler: what matters is how far apart samples are relative to a species' autocorrelation. A sampling effort E puts samples roughly $\sqrt{(10,000/E)}$ km apart, so the thresholds above can equivalently be stated as a ratio of sample spacing to length-scale: $d/\rho \approx 0.1$ for smoothly distributed species ($E \approx 4\text{--}5$ with $\rho \approx 500$ km), but $d/\rho \approx 0.6\text{--}0.7$ for patchily distributed ones ($E \approx 8\text{--}11$ with $\rho \approx 50$ km). The same reasoning should therefore carry to other systems and scales, though the ratio is not a single fixed number, because error depends on E and ρ jointly (Eq. 6).

Case study: agreement with observed data and implications

In the case study, the error model estimated from the simulations reproduced the structure of the model estimated from the observed data: the intercept (ω), the effect of sampling density (β), the effect of spatial autocorrelation (θ) and their interaction (γ) showed consistent directional effects and aligned closely with the 1:1 line (Appendix S1: Figure S1). This supports the use of the simulation-derived coefficients for taxa and domains similar to those studied here. Where the observed errors departed from the simulations, notably at intermediate autocorrelation levels, the

likely cause is the departure of real eDNA fields from a single stationary length-scale (see Assumptions and limitations of the method). Used with the species-specific length-scales, the planner showed that the 358 samples collected in 2019 met a target error of 0.5 only for the smoothest taxa, whereas taxa with $\rho < 210$ km would have required roughly two to twenty times as many samples (Table 1).

Applying our simulation-derived thresholds to our empirical data further illustrates this context dependency. Extrapolating the findings from the simulations, species (with high spatial autocorrelations) such as *Leuroglossus stilbius* and *Microstomus pacificus* would require only 4 samples per 10,000 km² to achieve a negligible marginal gain in precision (defined here arbitrary as $\Delta\epsilon = -0.05$). In contrast such as *Trachurus symmetricus*, *Thaleichthys pacificus*, *Stenobranchius leucopsarus*, and *Tarletonbeania crenularis* (species with low spatial autocorrelation) would require approximately 9 samples per 10,000 km² to reach the same relative threshold (Liu et al., 2026). The remaining species fall within the range of 4-9 samples per 10,000 km² (for context this is slightly higher than our current empirical grid and CalCOFI's standard 75-station grid with 3.6 and 3.4 samples per 10000 km² respectively), consistent with their intermediate spatial autocorrelation values. However, given that our analysis is based on 12 fish species, extending this conclusion to other taxa or ecosystems should be done with caution, and additional species may exhibit different scaling relationships.

~~Although we report effort per 10,000 km², the underlying point is simpler: what matters is how far apart samples are relative to a species' autocorrelation. A sampling effort E puts samples roughly $\sqrt{(10,000/E)}$ km apart, so the thresholds above can equivalently be stated as a ratio of sample spacing to length-scale: $d/\rho \approx 0.1$ for smoothly distributed species ($E \approx 4-5$ with $\rho \approx 500$ km), but $d/\rho \approx 0.6-0.7$ for patchily distributed ones ($E \approx 8-11$ with $\rho \approx 50$ km). The same reasoning should~~

~~therefore carry to other systems and scales, though the ratio is not a single fixed number, because error depends on E and ρ jointly (Eq. 6).~~

Assumptions and limitations of the method

Extrapolating directly from simulations to natural systems requires care because real eDNA fields may violate the simplifying assumptions of the stationary, isotropic squared exponential kernel. Spatial structure may vary across the domain (non-stationarity), differ by direction (anisotropy), or include sharp transitions (fronts, plume boundaries, shelf-break structure) that are not well represented by a globally smooth covariance function (Paciorek & Schervish, 2006). In addition, eDNA observation error is unlikely to be purely Gaussian, with variability arising from sources such as PCR bias, compositional constraints, incomplete calibration, and heterogeneity in filtration or extraction efficiency (Bessey et al., 2020; Gloor et al., 2017; Kelly et al., 2019; Shaffer et al., 2025). The modelling framework used here accounts for these sources (Guri et al., 2025), so they are unlikely to bias the results presented here. Finally, ecological processes such as migration, schooling, habitat discontinuities, or patchy prey fields can generate multi-scale patterns that depart from a single characteristic length-scale (Wagner & Fortin, 2005). These features may help explain elevated prediction error at intermediate autocorrelation levels in the empirical data relative to simulations, where additional sources of spatial complexity are not represented. Furthermore, the simulation assumes a stationary GP with fixed amplitude and noise and a single length-scale across depths, whereas real eDNA fields may have variable variance structure, multiple characteristic scales, and dependence on environmental covariates such as temperature, salinity, chlorophyll, and bathymetry (Liu et al., 2026). Our prediction error (ϵ_{sim}^2) is also computed against a "complete" prediction from a 4,000-point subset (Z) rather than the full grid,

a limitation most visible in the secondary thinning comparison, where the reference field is constrained to $N = 358$ samples, making prediction error higher and more stochastic.

Intrinsically, eDNA integrates ecological information through space and across time via transport and decay (Brandão-Dias et al., 2023; Fukaya et al., 2021). This means that even if organisms are locally patchy, the observed eDNA field may be smoother than the organism field, depending on hydrodynamic conditions and degradation rates. In our framework, such integration is embedded within the inferred length-scale, reinforcing that ρ should be interpreted as an effective length scale of the observed eDNA field rather than a purely ecological niche parameter (Jo et al., 2025). Integrating eDNA observations with independent abundance data (trawl, acoustics) would allow some degree of separation of the spatial autocorrelation parameter, enabling clearer ecological interpretation of ρ and improving the transferability of sampling-effort recommendations (Guri et al., 2024).

Spatial scale also sets an important boundary on the applicability of these results. The relationships identified here are expected to hold most strongly when interpreted relative to the extent and resolution of the study domain, rather than as fixed thresholds transferable across all systems. In a small study domain, the autocorrelation scale of the observed eDNA field may exceed the survey extent, causing the field to appear nearly uniform and making μ_d difficult to estimate independently of the spatial process (Paciorek, 2010; Redfern et al., 2006).

Perspectives and applications

The method turns a single estimable quantity, the spatial autocorrelation of the target taxon, into a defensible sampling design, and the planner places that calculation in the hands of those who allocate ship time. For data collection, a survey can be designed before a vessel leaves port by combining a prior estimate of ρ with the precision a program requires, converting a fixed budget

of ship time into a station count or identifying the effort needed to reach a target precision. For monitoring and management, the same inversion shows where existing programs are over- or under-sampled relative to their objectives. For policy and reporting, the planner documents the assumptions (ρ , ε and calibration) behind a chosen effort, so that sampling requirements in monitoring plans, survey proposals and emerging eDNA reporting standards can be stated transparently and revisited as new data accumulate.

Sampling effort required for eDNA surveys should be matched to the intended inferential goal. When the objective is to estimate GP parameters such as the mean (μ_d) for deriving depth-specific or region-specific abundance indices typically used in stock assessment indices (Shelton et al., 2022), the autocorrelation (ρ) for applied ecological questions, like implementing protected area networks or monitoring programs (Laroche, 2024), or the variance parameter (α^2), random sampling designs that adequately cover the full spatial domain of the species spatial autocorrelation are likely to be sufficient (Rocha et al., 2021), as sampling effort appears to have relatively limited influence on parameter estimation (Figure 54; Appendix S1: Section S4). In contrast, when the objective shifts toward predicting spatial distributions - such as identifying hotspots, mapping species occurrence, or quantifying spatial niche breadth - the accuracy of these predictions is much more sensitive to both sampling effort and the underlying spatial autocorrelation of the species (Figure 43A). Prediction is the more demanding of the two because it requires both accurate covariance parameters (Eq. 1–2) and observed points close enough to the location being predicted (Eq. 3).

This distinction also clarifies how sampling design shapes inference. Estimation of the mean benefits primarily from coverage (sampling across the full domain) and secondarily from balance (avoiding over-sampling a limited region) (Rocha et al., 2021). By contrast, estimation of the

length-scale parameter requires not only broad coverage but also a wide distribution of pairwise distances, including closely spaced stations that resolve rapid decorrelation and widely spaced stations that constrain the long-distance tail of the covariance function (Kaufman & Shaby, 2013). In practice, the starting point is a clear statement of the inferential objective and the precision it demands, since this determines whether effort should prioritize spatial coverage or the resolution of fine-scale autocorrelation. This is complemented by a prior estimate of the target taxon's spatial autocorrelation, drawn from published geostatistical fits, a pilot survey, or the depth-related pattern reported here, which the planner uses (Eq. 6–7) to characterize the relationship between sampling density (E) and expected accuracy. From this relationship, a sampling density can be identified beyond which additional effort yields diminishing returns relative to the precision required. Because the length-scale parameter cannot be recovered from a uniformly thinned grid, the resulting design should also be assessed for its inclusion of both widely and closely spaced station pairs. In multispecies programmes, these considerations are most appropriately governed by the patchiest taxon the survey is intended to resolve.

Applied to existing programs this yields a concrete conclusion. The West Coast eDNA grid analyzed here (3.6 samples per 10,000 km² per depth layer) and the CalCOFI 75-station grid (3.4 samples per 10,000 km²) already sit close to the effort at which returns diminish for smoothly distributed species and for estimating depth-specific means, so programs whose primary product is an abundance index need not expand. Programs intended to map patchily distributed taxa, including several forage and mesopelagic species examined here, are under-sampled by a factor of roughly two to three, and that shortfall is spatial rather than analytical (it cannot be recovered by sequencing more deeply or filtering more water at stations already occupied).

For managers and survey designers the practical message is that effort should be set by the patchiness of the taxa a programme is required to resolve and by the decision the data must support, and that both can be specified in advance rather than discovered afterwards. Nevertheless, across the scenarios considered here and the spatial domain used in this study, sampling effort of 4 - 9 samples per 10,000 km² per depth layer marked the point at which diminishing returns became apparent given species-specific spatial autocorrelation. We therefore recommend that eDNA surveys report estimated length-scales alongside detection or concentration results; as such estimates accumulate, taxon- and region-specific priors will make prospective design routine rather than bespoke, and recalibrated coefficients can be entered directly into the planner.

Acknowledgments

This material is based upon research supported by the Office of Naval Research under Award Number N00014-22-1-2719 and support from the National Marine Fisheries Service's Genomic Strategic Initiative. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Author contributions

Gledis Guri conceived the study, developed the methodology and the survey planner, performed the analyses and wrote the manuscript; Andrew O. Shelton contributed to the methodology, funding acquisition, data interpretation and discussion; Ryan P. Kelly contributed to conceptualization, funding acquisition, data interpretation and discussion; Owen R. Liu contributed to data interpretation and discussion; Krista M. Nichols contributed to conceptualization, funding acquisition and discussion. All authors contributed critically to the drafts and gave final approval for publication.

Conflict of interest statement

The authors declare no conflicts of interest.

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Figure captions

Figure 1. Conceptual illustration of how sampling density affects the reconstruction of (A) a patchy and (B) a smooth concentration field along a one-dimensional transect (km). Top rows show the true concentration (black line); middle and bottom rows show reconstructions (red dashed line) from few and many samples (green circles), respectively.

Figure 2. Workflow of the method for simulated (A) and empirical (B) data. (A) A field generated from known parameters on a grid (Q , $N = 18,916$) is sampled (Y , $N = 4,000$) to give a reference prediction (Z); ε_{2sim} compares Z with predictions from thinned designs (Y_E), and $\varepsilon_{3sim,358}$ uses the prediction from $N = 358$ samples ($S_{E=358}$) as the reference. (B) The same thinning applied to the observed data ($N = 358$) quantifies ε_{4obs} .

~~**Figure 3.** Gaussian process parameter estimates for the 12 species. (A) Spatial autocorrelation length scale (ρ , km) by depth association (surface, midwater, deepwater). (B) Depth-specific mean eDNA concentration (μ ; log copies/L). (C) Amplitude (α^2) versus non-spatial variance (σ^2). Points are posterior means and bars 95% credible intervals; colors identify species.~~

Figure 43. (A) Prediction error (ε_{2sim}) across sampling densities (E , samples per 10,000 km² per depth layer) for simulated fields with different length-scales (colors); points are simulation outcomes and lines mean trends. (B) Marginal change in error ($\Delta\varepsilon_{2sim}$) per unit increase in E , predicted from Eq. 6 (Appendix S1: Figure S5); the dashed line marks the threshold ($\Delta\varepsilon_{2sim}/\Delta E = -0.05$) for negligible improvement.

Figure 54. Recovery of Gaussian process parameters in the simulations: mean root mean squared error (τ) between estimated and true values across 20 replicates for (A) ρ , (B) σ^2 , (C) depth-specific mean (μ_d) and (D) α^2 , by sampling density (E) and length-scale (colors).

Figure 5. Gaussian process parameter estimates for the 12 species. (A) Spatial autocorrelation length-scale (ρ , km) by depth association (surface, midwater, deepwater). (B) Depth-specific mean eDNA concentration (μ ; log copies/L). (C) Amplitude (α^2) versus non-spatial variance (σ^2). Points are posterior means and bars 95% credible intervals; colors identify species.

Figure 1

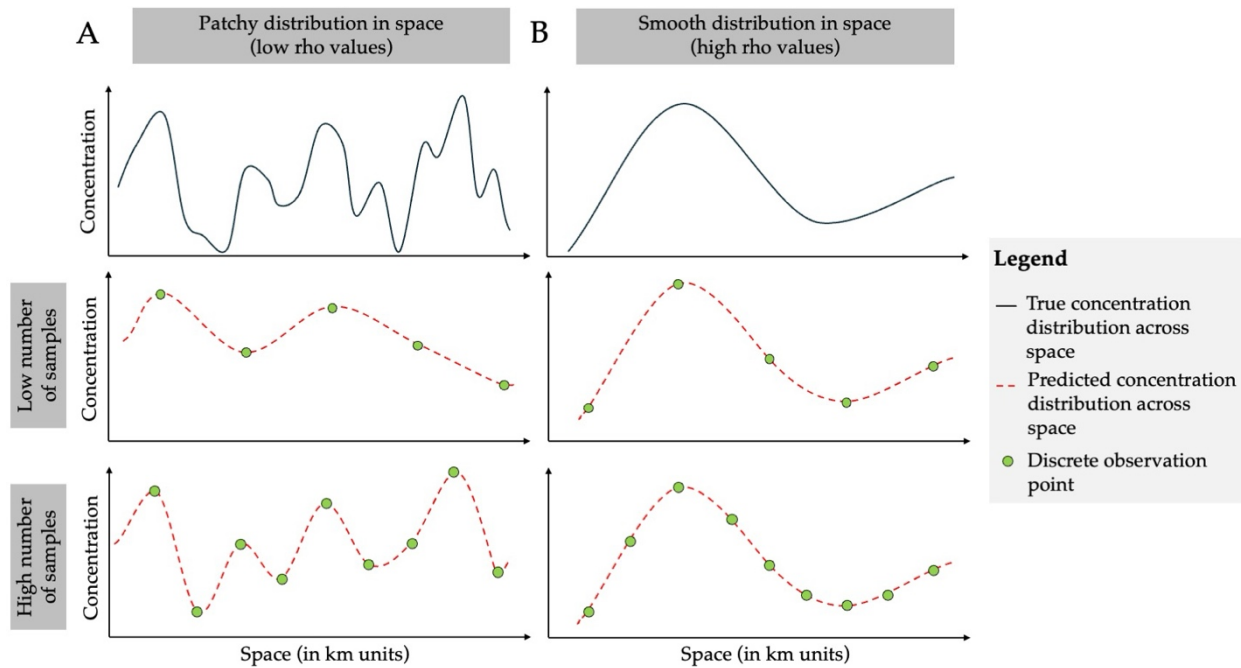


Figure 2

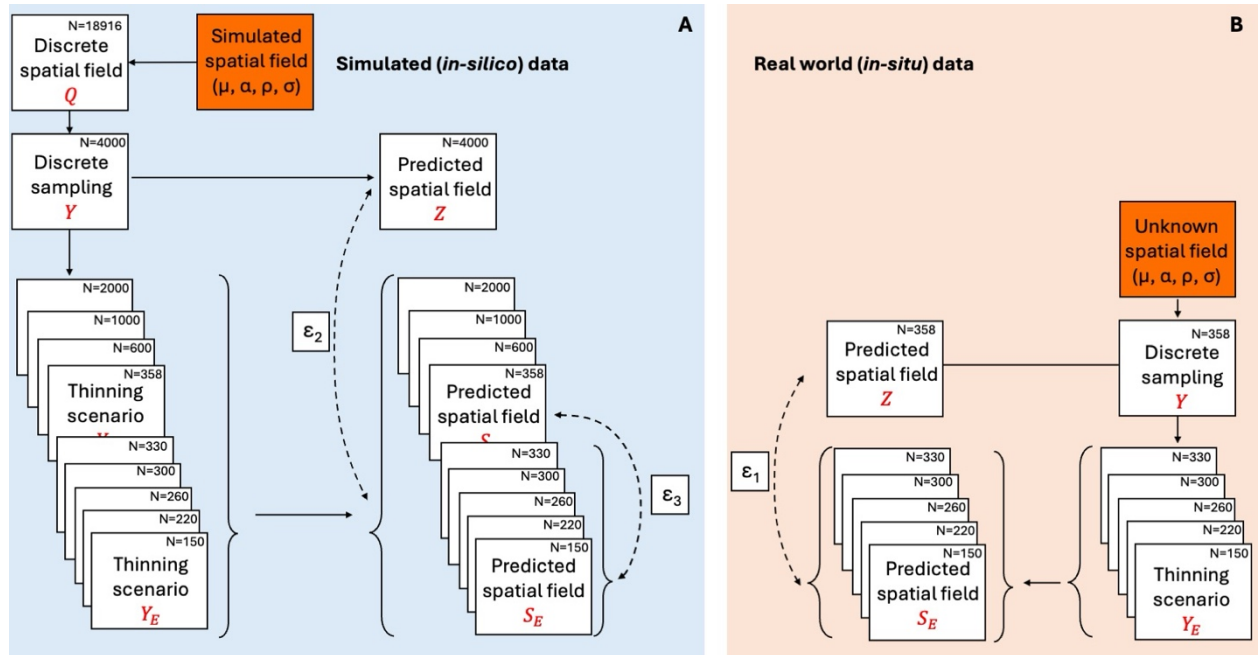
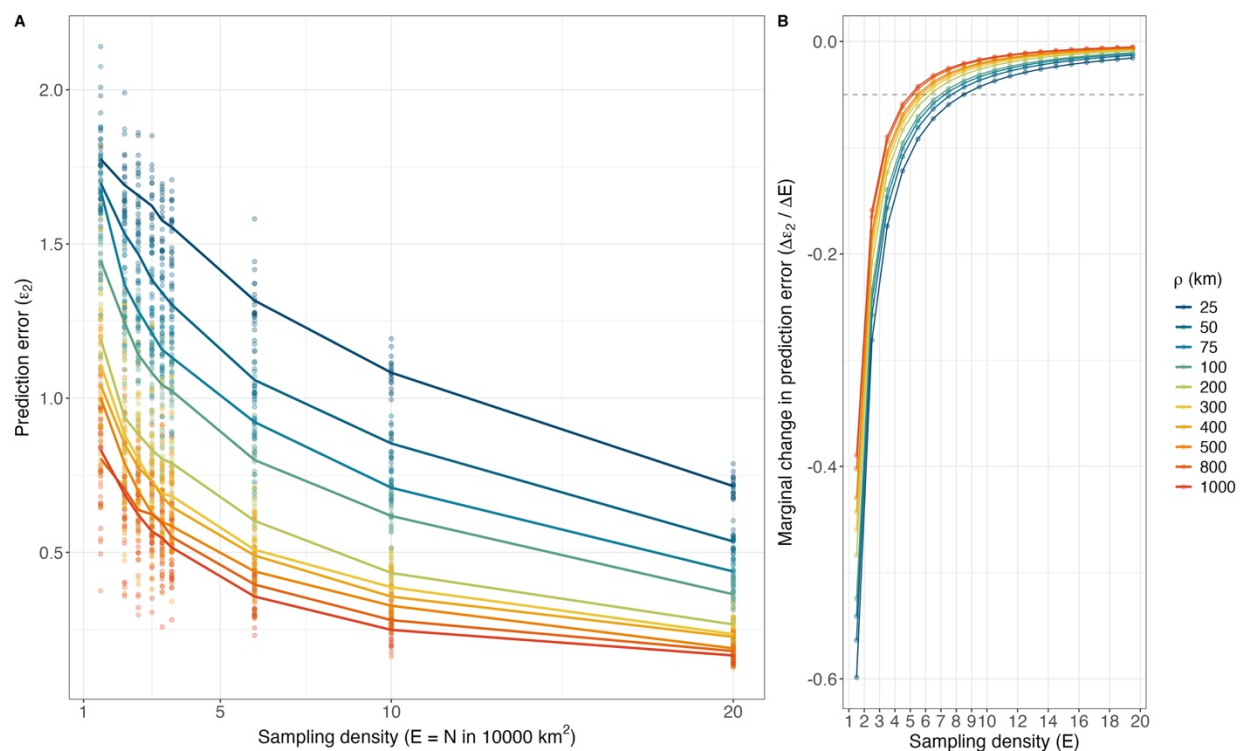
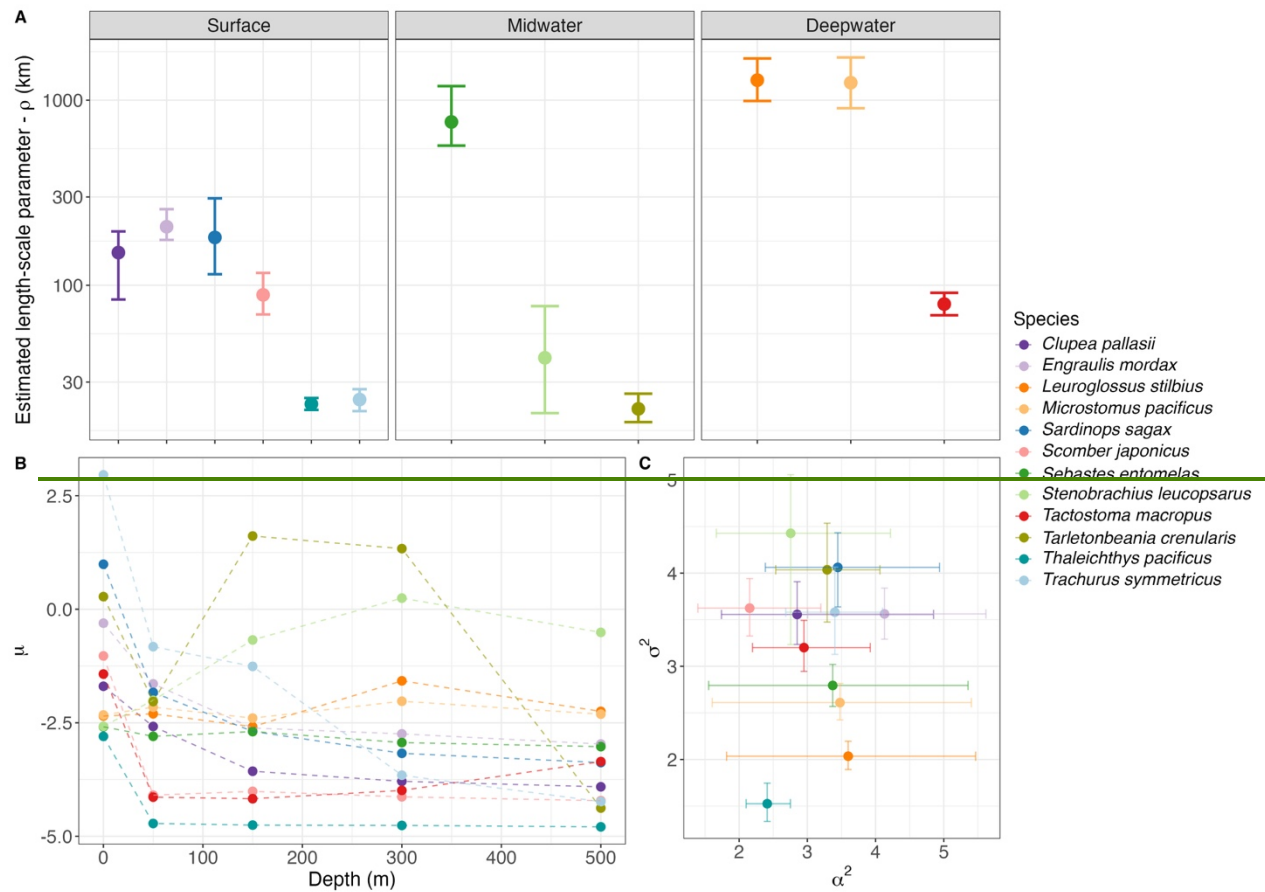
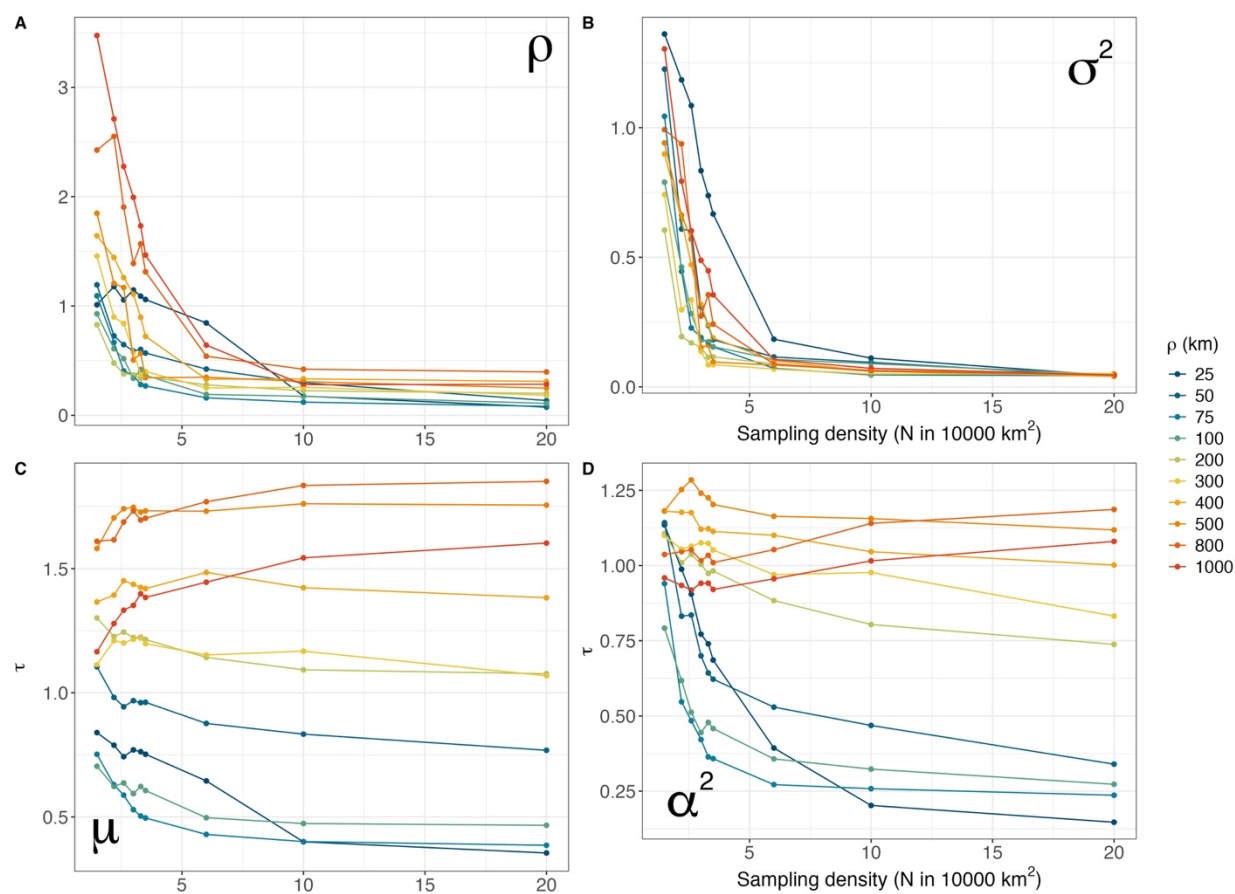
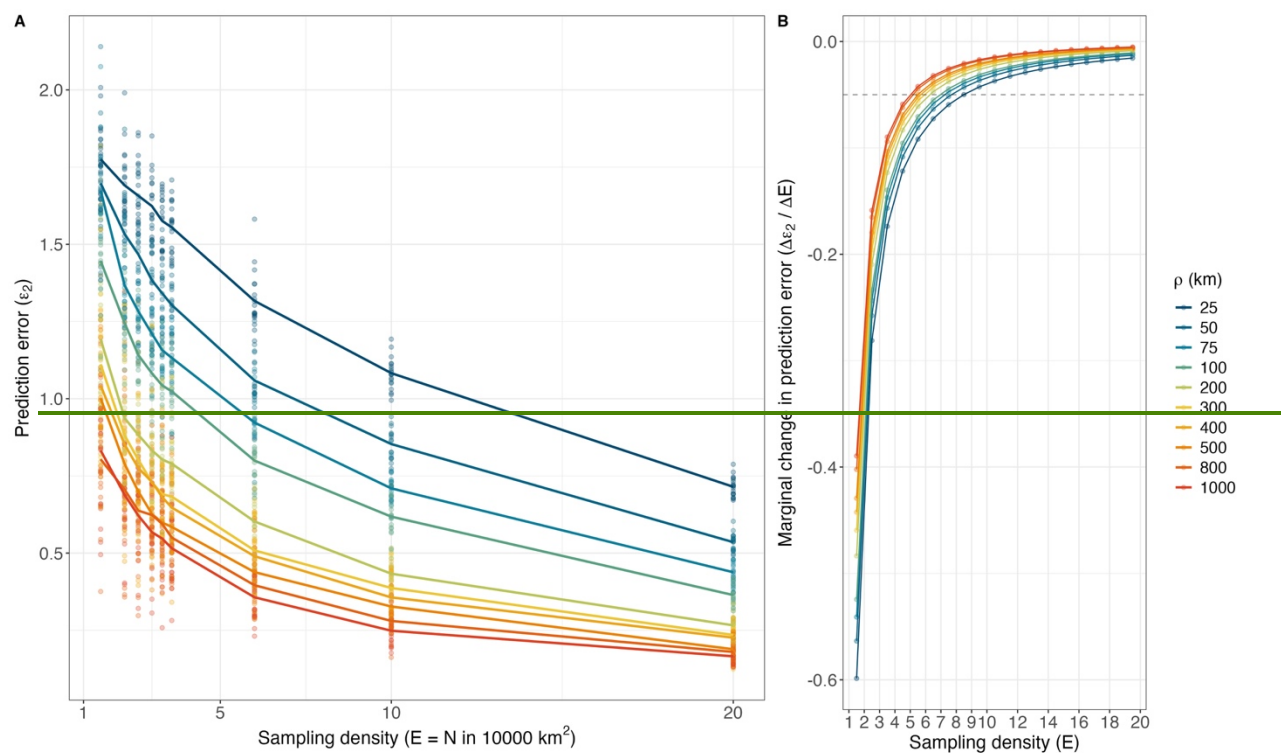


Figure 3



743 Figure 4



746 Figure 5

